Data Mining Assignment on Logistic Regression

1 Learning Goals

By completing this assignment, students should be able to:

- Interpret logistic regression coefficients biologically
- Explain odds and odds ratios
- Compute probabilities from log-odds
- Implement logistic regression in Python
- Evaluate classifiers using ROC and PR curves
- Understand the effect of class imbalance

2 Scenario Context

You will be studying a binary phenotype:

- $Y = 1$: Disease present
- $Y = 0$: Disease absent

Features may include:

- Gene expression levels
- Clinical covariates (age, BMI)

3 Problems

3.1 Problem 1: Conceptual Foundations

(A) Explain the meaning of:

$$\ln\left(\frac{p}{1-p}\right)$$

in biological terms.

(B) If a coefficient $\beta_1 = 0.7$, interpret:

(a) The effect on log-odds.
(b) The odds ratio.
(c) The biological meaning.

(C) Why do probabilities exhibit diminishing returns near 0 and 1 in logistic regression?

3.2 Problem 2: Manual Computation

Suppose we have a fitted model:

$$\ln\left(\frac{p}{1-p}\right) = -2 + 0.8 \cdot \text{GeneA}$$

(A) Compute the probability of disease when GeneA $= 0$.

(B) Compute the probability when GeneA $= 3$.

(C) What is the odds ratio for a 1-unit increase in GeneA?

(D) Interpret the biological implication.

3.3 Problem 3: Python Implementation

(A) Generate synthetic gene expression data using

```
from sklearn.datasets import make_classification
```

Specifications for `make_classification` are shown in Figure 1. Here, the number of features $d = 20$.

```
my_class_sep = 1

# Generate synthetic data
X, y = make_classification(
    n_samples=2000,
    n_features=20,
    n_informative=5,
    n_redundant=2,
    n_repeated=0,
    n_classes=2,
    weights=[0.8, 0.2], # [0
    class_sep=my_class_sep,
    random_state=42
)
```

Figure 1: Specifications for creating the synthetic dataset.

(B) Split the data into training dataset and testing dataset by using:

```
 X_train, X_test, y_train, y_test =
train_test_split( X, y, test_size=0.3, stratify=y, random_state=42)
```

(C) Set up a pipeline using `sklearn.pipeline` that will standardize the training dataset and then fit logistic regression.

(D) Report the logistic regression model that you have trained:

(a) Make a plot of β_j for $j = 1, 2, \cdots, d$. Include a flat line of $\beta_j = 0$ for all j in the plot. Your figure should look something like Figure 2.

(b) What can we say about those $\beta_i > 0$?

(c) Make a plot of e^{β_j} for $j = 1, 2, \cdots, d$. Include a flat line of $e^{\beta_j} = 1$ for all j in the plot. Your figure should look something like Figure 3.

(d) What can we say about those odds that are greater than 1?

(e) What is the value of the intercept?

(E) Predicted probabilities

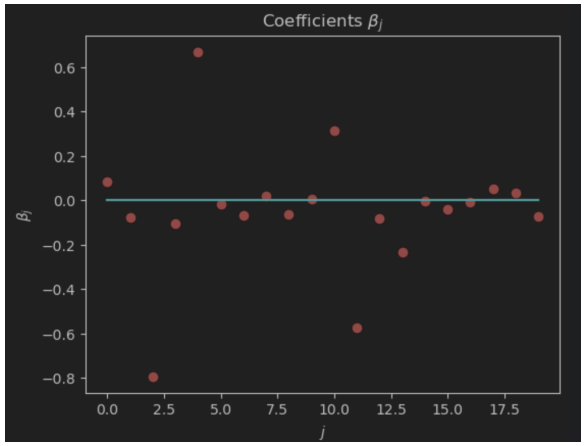

Figure 2: β_j vs j.

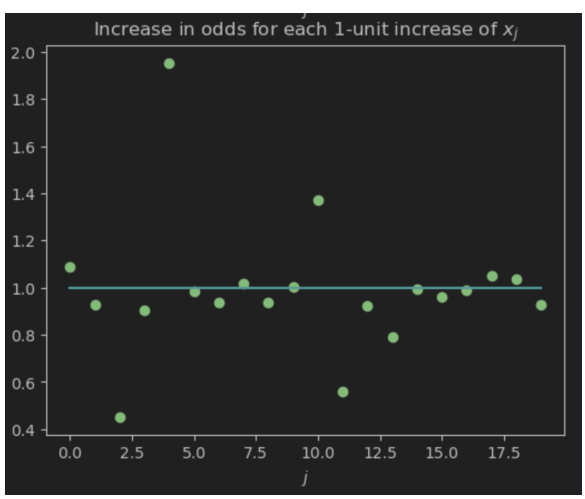

Figure 3: Odds increase vs j.

(a) Using the pipeline that you have set up earlier to standardize X_{test} (i.e., use the mean and standard deviation of the training data to standardize the testing data to prevent data leakage from happening) and then predict the probability $P(Y = 1|X)$ for each test sample using the standardized testing data. Make a plot of the predicted probabilities for all of the test samples. Include a flat line of the predicted probabilities = 0.5 in the plot. Your plot should look something like Figure 4.

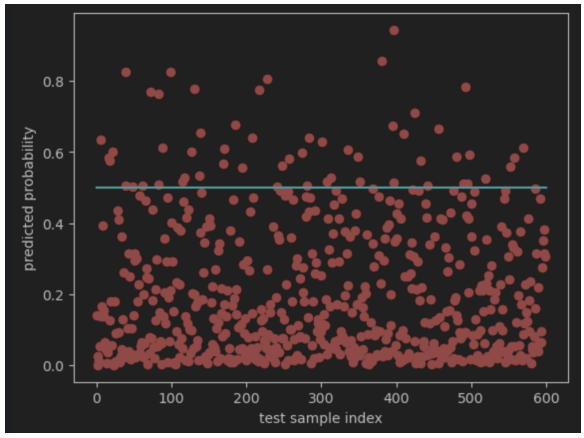

Figure 4: Predicted probability vs sample index.

(b) Let the probability threshold be 0.5, 0.7, and 0.9, respectively, for predicting $y = 1$ for each test sample. For each probability threshold, print out the confusion matrix, False Positive Rate (FPR), and True Positive Rate (TPR). Describe how FPR and TPR change as the probability threhold increases.

(F) Plot ROC curve and compute AUC.

(G) Plot Precision-Recall curve and compute the average precision score.

(H) Your modeling and analysis above used data generated by `make_classification` with my_class_sep = 1 (Figure 1). Change `my_class_sep` to to 2 and then 3, respectively, plot the ROC and PR curves. Describe how the AUC and PR curves change when `my_class_sep` increases from 1 to 2 to 3. Explain why you seen this change.

(I) Explain why ROC may look good but PR may look poor in imbalanced datasets.

4 Answers

4.1 Problem 1: Conceptual Foundations

(A) The expression

$$\ln\left(\frac{p}{1-p}\right)$$

is the log-odds of disease where

$$\text{odds} = \frac{p}{1-p}$$

(B) If $\beta_1 = 0.7$:

- log-odds increase by 0.7 per 1 unit increase in feature x_1.
- The odds ratio $= e^{0.7} \approx 2.01$. So the odds roughly double.
- biological meaning: 1 unit increase in the feature is associated with $\sim 2\times$ higher odds (holding others fixed).
- Sigmoid saturates near 0 and 1; equal changes in log-odds produce smaller changes in probability near the extremes.

4.2 Problem 2: Manual Computation

Model:

$$\ln\left(\frac{p}{1-p}\right) = -2 + 0.8 \cdot \text{GeneA}$$

(A) When GeneA = 0:

$$\text{logit} = -2$$

so,

$$p = \frac{1}{1+e^2} \approx 0.119$$

(B) When GeneA = 3:

$$\text{logit} = -2 + 2.4 = 0.4$$

so,

$$p = \frac{1}{1+e^{-0.4}} \approx 0.599$$

(C) Odds-ration for 1-unit increase in GeneA:

$$e^{0.8} \approx 2.23$$

(D) Biological interpretation: Each 1-unit increase of GeneA multiplies odds by $\sim$2.23 (holding others fixed).

Note: Check the Python Notebook for verification of the manual computations.

4.3 Problem 3: Python Implementation

From the figures, we can see that when `class_sep` increases, the AUC and AP both improve. This is because differences of data for the two classes increase as `class_sep` increases.

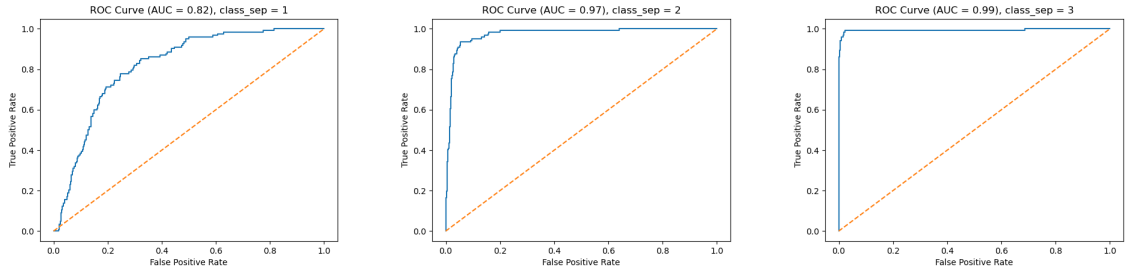

Figure 5: ROC curves for class_sep = 1 (Left), 2 (Middle), and 3 (Right)

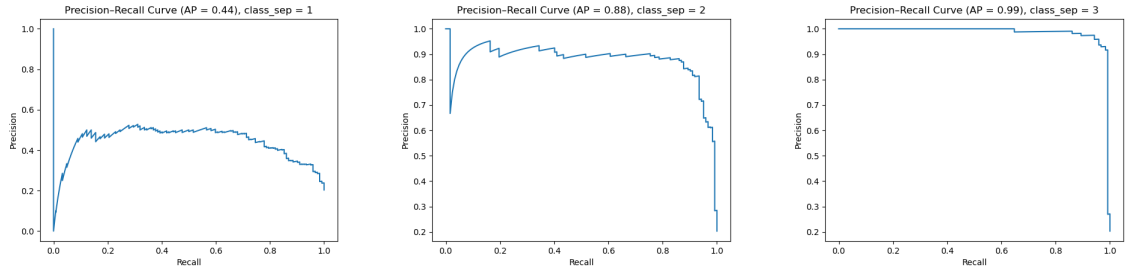

Figure 6: PR curves for class_sep = 1 (Left), 2 (Middle), and 3 (Right)